Tables

1. The Master Data (Your Ground Truth)
omcs: This is the master list of your clients (Oil Marketing Companies). Without this, you cannot track who is responsible for the leaked revenue. It serves as the central hub connecting physical dispatches to financial payments.

tariffs: This holds the dynamic pricing rules for pipeline transport and storage. By separating tariffs from the transactional data, you can prove to the judges that your pipeline accurately calculates expected revenue based on specific effective dates, rather than hardcoded assumptions.

2. The Physical vs. Commercial Core (Finding Unbilled Revenue)
depot_loading_logs: This represents the physical reality—what the metering system recorded at the loading bay before the truck drove off.

dispatches: This represents the commercial reality—the official waybill that KPC intends to bill for.

Relevance: By reconciling depot_loading_logs against dispatches, you can identify "unmetered loading" (product that left the depot but never got a commercial waybill). This directly addresses the requirement to reconcile product loading and dispatch events.

3. The Financial Flow (Finding Uncollected Revenue)
invoices: What KPC officially billed the OMCs.

payments: The actual cash KPC received in the bank.

Relevance: By reconciling dispatches against invoices, you find "unbilled leakage" (dispatches that were forgotten and never invoiced). By reconciling invoices against payments, you identify underpayments, installments, or entirely unpaid invoices.

4. Operational & Technical Governance (Scoring High on the Rubric)
depot_daily_inventory: This provides daily snapshots of physical tank dips versus book balances. It allows you to detect macro-level product shrinkage or physical leakage before the product even reaches the loading gantry.

quarantine_audit_log: This is your silver bullet for the rubric's "Technical quality (25%)" criteria. Instead of silently deleting bad data (like missing dates or duplicate records), your ETL pipeline safely quarantines them here. When the judges ask for QA evidence, you will show them this table to prove your pipeline handles real-world operational messiness safely.

Columns

Here are the concise descriptions for every column across all eight tables. You can use these as a data dictionary for your hackathon documentation.

1. omcs (Oil Marketing Companies Master)

omc_id: Unique identifier for the Oil Marketing Company.

customer_name: The registered business name of the OMC.

kra_pin: Kenya Revenue Authority tax identification number.

payment_terms_days: The agreed number of days the OMC has to settle invoices (e.g., 15, 30).

credit_limit_kes: Maximum allowable credit limit in Kenyan Shillings.

risk_rating: Assessed financial risk level (Low, Medium, High).

contact_email: Primary email address for billing.

phone: Main contact phone number.

is_active: Boolean indicating if the OMC is currently permitted to load fuel.

2. tariffs (Pricing Master)

tariff_id: Unique identifier for a specific tariff rate version.

product: The petroleum product the rates apply to (e.g., Petrol, Jet A-1).

depot: The specific KPC facility the rates apply to.

pipeline_tariff_per_m3_km: The transport cost per cubic meter per kilometer.

storage_tariff_per_m3_day: The storage cost per cubic meter per day.

effective_start: The date these specific rates became active.

effective_end: The date these specific rates expired.

3. depot_loading_logs (Physical Meter Events)

loading_id: Unique identifier for the physical loading event.

gantry_bay_id: The specific physical bay where the truck was loaded.

meter_id: The ID of the flow meter that recorded the volume.

dispatch_id: The commercial waybill linked to this loading (may be empty if bypassed).

omc_id: The OMC receiving the product.

depot: The location of the loading gantry.

product: The specific fuel type loaded.

physical_loaded_liters: The actual raw volume dispensed by the meter.

temperature_c: The temperature of the fuel during loading (affects volume expansion).

density_kg_m3: The physical density of the product at the time of loading.

loading_timestamp: The exact date and time the truck was filled.

4. dispatches (Commercial Waybills)

dispatch_id: Unique identifier for the commercial waybill.

loading_id: Link back to the physical loading meter record.

date: The commercial date the dispatch was recorded.

year: Extracted year for easy dashboard aggregations.

month: Extracted month for easy dashboard aggregations.

omc_id: The OMC responsible for the dispatch.

customer_name: Name of the OMC.

product: The fuel type dispatched.

depot: The originating KPC depot.

volume_liters: The commercially declared volume (should match the physical loading).

distance_km: The transport distance the OMC is being billed for.

transport_tariff_kes: The calculated transport fee in KES.

storage_tariff_kes: The calculated storage fee in KES.

value_kes: The total commercial value of the dispatch (transport + storage).

5. invoices (Financial Billing)

invoice_id: Unique identifier for the financial invoice.

dispatch_id: The specific dispatch this invoice is billing for.

omc_id: The OMC being billed.

customer_name: Name of the OMC.

product: The product being billed.

date: The date the invoice was officially issued.

value_kes: The total financial amount billed to the OMC.

6. payments (Financial Collections)

payment_id: Unique identifier for the payment receipt.

invoice_id: The specific invoice this payment is settling.

omc_id: The OMC making the payment.

customer_name: Name of the OMC.

date: The date the payment was received by KPC.

value_kes: The actual monetary amount paid.

installment_no: Indicates if the payment is partial (e.g., 1 for the first installment, 2 for the second).

7. depot_daily_inventory (Physical Stock Reconciliation)

record_id: Unique identifier for the daily inventory log.

date: The day the inventory check was performed.

depot: The storage facility location.

product: The type of fuel being measured.

opening_stock_liters: Volume in the tank at the start of the day.

received_pipeline_liters: Volume pumped into the tank from the pipeline that day.

dispatched_liters: Total volume officially loaded out to trucks that day.

book_closing_stock_liters: The mathematical expected volume at the end of the day (Opening + Received - Dispatched).

physical_dip_closing_stock_liters: The actual volume measured by physically dipping the tank.

variance_liters: The difference between the book value and the physical dip (flags unexplained loss/gain).

8. quarantine_audit_log (ETL Data Quality Evidence)

quarantine_id: Unique identifier for the rejected data row.

dataset_name: The source table where the error occurred (e.g., 'invoices').

failed_gate: The specific ETL quality gate that rejected the row (e.g., Gate C - Date Standardizer).

reason: Text description of why it failed (e.g., 'Missing/Unparseable date').

quarantined_at: Timestamp of exactly when the pipeline caught and isolated the error.

raw_record_json: The original, corrupted row data preserved in JSON format for debugging and compliance.